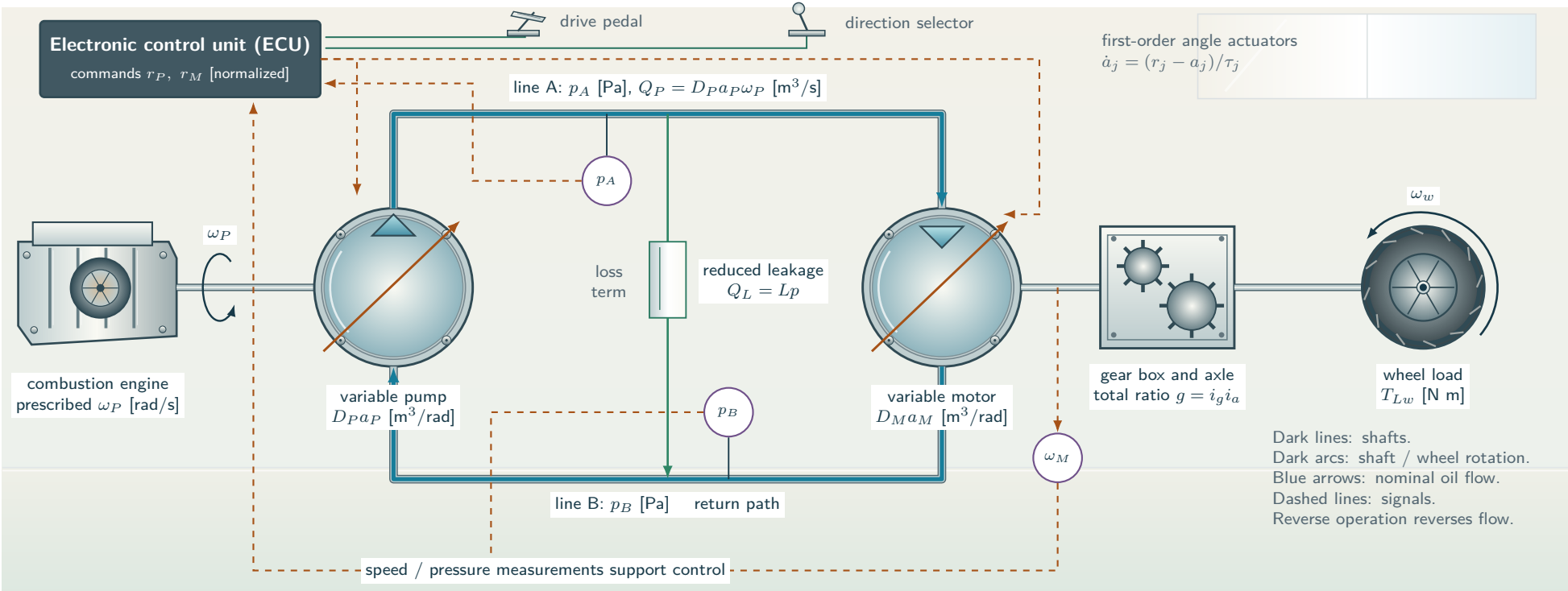
Hydrostatic drive: closed oil circuit and vehicle load

Paper section 3.6: *Hydrostatic transmission in mobile working machines*

Variable-displacement pump and motor couple hydraulic pressure to shaft speed



SI variant state and storage

$[a_P, a_M, p, \omega_M]^\mathsf{T}$, $p = p_A - p_B$.
$C_\Delta \dot{p} = D_P a_P \omega_P - D_M a_M \omega_M - Lp$.
C_Δ [m^3/Pa]: effective capacitance.

Mechanical balance

$J\dot{\omega}_M = \eta D_M a_M p - b\omega_M - T_{Lw}/g$.
Motor torque uses motor displacement.
Displacement per revolution is converted through $D = V/(2\pi)$.

Declared interpretation

Eq. (35) remains a separate transcription. The SI variant requires explicit displacement and capacitance conventions. No electrical actuator calibration is inferred.

Functional circuit, not a standardized hydraulic drawing. Constant efficiencies are valid for the declared motoring study; no passive four-quadrant, cavitation or relief-valve model is implied. See HYDROSTATIC_SOURCE_REVIEW.md for source conflicts and unresolved calibration.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.